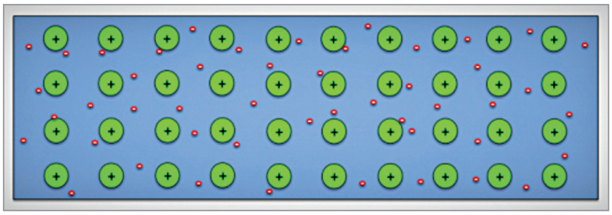

Fig. 2: Uniform distribution of neutral atoms

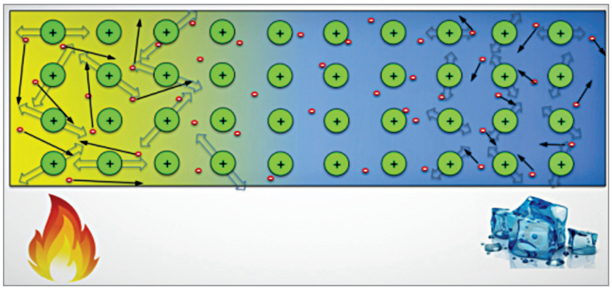

Fig. 3: Generation of potential difference due to heating in TEG

which is now leading to a quantum jump in solar energy usage.

Photovoltaics

A photovoltaic cell is made of semiconductor materials that absorb photons of the sunlight and generate a flow of electrons. Photons are elementary particles generated by Sun that carry solar radiation at a speed of 300,000km/s. When the photons strike a semiconductor material like silicon, they release the electrons from its atoms, leaving behind a vacant space called 'hole.' The stray electrons move around looking for a hole to fill.

Generally, a PV cell is made up of two types of silicon. The silicon wafer that is exposed to the Sun is doped with atoms of phosphorus, which has one more electron than silicon. The back side of the cell is made of silicon doped with atoms of boron, that has one less electron than silicon.

The sandwich thus constructed works like a battery. The layer that has surplus electrons becomes the negative terminal (n) and the other side that has a deficit of electrons acts as the positive terminal (p). An electric field is created between the two layers at the junction.

On excitation by photons electrons are swept to the n-side by the electric field at the junction, while the holes drift to the p-side. Both the sides are provided with metallic electrical contacts to collect electrons and holes. Electrons then flow in the external circuit in the form of electrical energy. Fig. 1 shows how a PV cell works.

Thermoelectricity

Thermoelectricity, as the name suggests, stands for conversion of thermal energy (temperature difference) into electricity. It encompasses mainly two phenomena: the Seebeck effect and Peltier effect.

Seebeck effect is the phenomenon that a potential difference will appear between the two ends of a metal or semiconductor wire when they are kept at different temperatures. The potential difference is proportional to the temperature difference and the material's property known as Seebeck coefficient.

All materials are made of atoms, and atoms contain positively charged nucleus with negatively charged electrons moving around them. The electrons that are closer to the nucleus are bound more strongly, whereas the outer ones are loosely bound. When the temperature is uniform, the distribution of negative electrons is uniform and neutralises positive ions everywhere in the material, as shown in Fig. 2.

But when one end of the wire is heated and the other end is kept cool, electrons at the hot end gain more energy and higher speed than those at the cool end, which is indicated by the longer arrows in Fig. 3. So, at any instant more electrons move to the cold end than those moving back. So, the hot end becomes positively charged and the cold end becomes negatively charged, and current flows through the external conductor of a thermoelectric generator (TEG).

A single thermoelectric device is constructed from two solid-state devices that are usually made from bismuth telluride (Bi_2Te_3), as shown in Fig. 4. One of these pellets of semiconductor is doped with acceptor impurity to create a p-type component to have more positive charged carriers or holes, thus providing a positive Seebeck coefficient. The other is doped with donor impurity to produce an n-type component to have more negative charged carriers, thus providing a negative type of Seebeck coefficient. The two semiconductor components are then physically connected serially on one side, usually with a copper strip, and mounted between two ceramic outer plates that provide electric isolation and structural integrity.

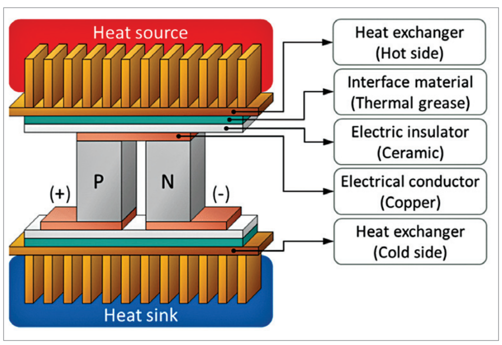

Fig. 4: Thermoelectric generator (TEG)

The Seebeck effect is a direct energy conversion of heat into a voltage potential. It occurs due to the movement of charge carriers within the semiconductor. Charge carriers diffuse away from the hot side of the semiconductor. This diffusion leads to a build-up of charge carriers at one end. This build-up of charge creates